

A candidate counterexample to Conjecture 24 in Infinite induced-saturated graphs

github.com/graph-theory-AI — machine-generated candidate writeup

17 September 2026

Abstract

The five-vertex tournament $C3[TT2,TT2,1]$ is a counterexample: every countable tournament avoiding it has a distinct locally finite perturbation that still avoids it. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Conjecture 24](#) from *Infinite induced-saturated graphs*.

1.1 Problem statement

Conjecture. Let T be a finite tournament. If T is not transitive, then there exists a countably infinite T -free tournament S_T such that every locally finite perturbation of S_T has an induced copy of T .

1.2 Catalog context

This is proposed as the tournament analogue of the paper’s main Theorem 1. The non-transitivity condition is necessary because any tournament on 2^n elements contains a transitive subtournament on n elements, so a transitive T cannot be avoided in an infinite structure under locally finite perturbations.

1.3 Claimed result

Model verdict: disproved (confidence: high).

The five-vertex tournament $C3[TT2,TT2,1]$ is a counterexample: every countable tournament avoiding it has a distinct locally finite perturbation that still avoids it.

The model’s own caveat: Perturbations are understood to be nonempty; the argument does not settle the weaker single-arc saturation question.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 1. The counterexample

Let T_* be the five-vertex tournament with vertex partition

$$A = \{a_0, a_1\}, \quad B = \{b_0, b_1\}, \quad \{c\},$$

and orientations

$$a_0 \rightarrow a_1, \quad b_0 \rightarrow b_1, \quad A \rightarrow B \rightarrow \{c\} \rightarrow A.$$

Thus

$$T_* = C_3[\text{TT}_2, \text{TT}_2, 1].$$

Here, an arrow between sets means that all arcs between those sets have the indicated direction.

This tournament is nontransitive. I will prove the following, which directly contradicts the conjecture.

2.1.1 Theorem

For every countably infinite T_* -free tournament S , there is a T_* -free tournament $S' \neq S$ on the same vertex set such that the pairs whose orientations differ between S and S' form a locally finite graph.

The key structural fact is stronger than needed:

Every countable T_* -free tournament can be made transitive by a locally finite set of arc reversals.

No result from the previous attempt is needed below.

2.2 2. A criterion for locally finite distance from a transitive tournament

For a tournament S , define an undirected graph $J(S)$ on $V(S)$ by declaring xy to be an edge precisely when the arc between x and y belongs to infinitely many directed triangles.

Write

$$O(x) = N_S^+(x).$$

For an arc $x \rightarrow y$, its directed-triangle completions are exactly

$$Z(x, y) = \{z : y \rightarrow z \rightarrow x\}.$$

In particular,

$$O(y) \setminus O(x) = Z(x, y). \quad (1)$$

2.2.1 Lemma 1

A countable tournament S differs from some transitive tournament on $V(S)$ by a locally finite set of arc reversals if and only if $J(S)$ is locally finite.

2.2.1.1 Proof Necessity. Suppose L is transitive and the change graph $D = D(S, L)$ is locally finite.

If $xy \notin E(D)$, then every directed triangle xyz of S must have $xz \in E(D)$ or $yz \in E(D)$: otherwise that triangle would also occur in L . Thus there are only finitely many directed triangles of S containing xy . Consequently,

$$J(S) \subseteq D,$$

so $J(S)$ is locally finite.

Sufficiency. Suppose $J(S)$ is locally finite. For sets X, Y , write

$$X \subseteq^* Y \iff X \setminus Y \text{ is finite.}$$

Define an equivalence relation on $V(S)$ by

$$x \sim y \iff O(x) \triangle O(y) \text{ is finite.}$$

On the equivalence classes, reverse almost-inclusion of outneighbourhoods defines a partial order: a class C precedes a distinct class D when

$$O(y) \subseteq^* O(x) \quad (x \in C, y \in D).$$

This is well-defined, and antisymmetry follows from the definition of $\sim$. Choose a linear extension of this partial order.

If $x \rightarrow y$ and $xy \notin E(J(S))$, then (1) gives

$$O(y) \subseteq^* O(x).$$

Therefore, if x and y belong to different equivalence classes, their original orientation agrees with the chosen class order. Hence:

$$\text{Every incorrectly oriented pair between classes belongs to } J(S). \quad (2)$$

It remains to order each equivalence class with only locally finite discrepancies.

Fix a class C , choose $r \in C$, and partition it as

$$C = C^- \cup \{r\} \cup C^+,$$

where

$$C^- = \{x \in C : x \rightarrow r\}, \quad C^+ = \{x \in C : r \rightarrow x\}.$$

For every $x \in C$, the sets $O(x)$ and $O(r)$ differ finitely. It follows that:

- every vertex of C^- has finite outdegree within C^- ;
- every vertex of C^+ has finite indegree within C^+ ;
- the pairs disagreeing with the orientation $C^- \rightarrow C^+$ form a locally finite graph.

Enumerate C^- and C^+ , using finite enumerations when appropriate. Order C^- in reverse enumeration order and C^+ in enumeration order, and concatenate them as

$$C^- < r < C^+. \quad (3)$$

To check local finiteness within C^- , consider its i -th enumerated vertex. Among vertices enumerated before it, there are only i possible discrepancies; among vertices enumerated after it, discrepancies are contained in its finite outneighbourhood in C^- . The analogous argument for C^+ uses finite indegree. Together with the third bullet above and the exact orientations through r , this proves that the discrepancies inside C are locally finite.

Finally, order all vertices first by the chosen class order and then by the orders (3). Let L be the resulting transitive tournament. By (2), its cross-class discrepancies with S lie in the locally finite graph $J(S)$; its within-class discrepancies are locally finite by construction. Each vertex therefore belongs to only finitely many changed pairs. $\square$

2.3 3. Avoiding T_* forces $J(S)$ to have maximum degree at most two

For an arc $x \rightarrow y$, also define

$$P(x, y) = \{p : x \rightarrow p \rightarrow y\}.$$

These are the vertices that would complete a directed triangle if xy were reversed.

2.3.1 Lemma 2

If S is T_* -free and $x \rightarrow y$ has $P(x, y) \neq \emptyset$, then

$$|Z(x, y)| \leq 2.$$

2.3.1.1 Proof Choose $p \in P(x, y)$. Suppose that $Z(x, y)$ contains three vertices. Among them, two, say z_1, z_2 , have the same orientation relative to p .

If $p \rightarrow z_1, z_2$, then

$$\{y, p\} \rightarrow \{z_1, z_2\} \rightarrow \{x\} \rightarrow \{y, p\}.$$

These five vertices induce T_* .

If $z_1, z_2 \rightarrow p$, then

$$\{z_1, z_2\} \rightarrow \{x, p\} \rightarrow \{y\} \rightarrow \{z_1, z_2\},$$

again inducing T_* .

In either case, the two-vertex blocks automatically induce transitive tournaments. Both possibilities contradict T_* -freeness. $\square$

2.3.2 Lemma 3

In any tournament, the arcs $x \rightarrow y$ satisfying

$$P(x, y) = \emptyset \quad (4)$$

form an undirected graph of maximum degree at most two.

2.3.2.1 Proof At any vertex x , at most one outgoing arc can satisfy (4). Indeed, if both $x \rightarrow y$ and $x \rightarrow z$ satisfied it, then one of $y \rightarrow z$ and $z \rightarrow y$ holds. In the first case,

$$x \rightarrow y \rightarrow z$$

contradicts $P(x, z) = \emptyset$; the second case similarly contradicts $P(x, y) = \emptyset$.

Likewise, at most one incoming arc at x can satisfy (4). Thus the corresponding undirected degree is at most two. $\square$

2.3.3 Corollary 4

For every T_* -free tournament S ,

$$\Delta(J(S)) \leq 2.$$

Consequently, every countable T_* -free tournament is a locally finite perturbation of a transitive tournament.

2.3.3.1 Proof If an arc belongs to infinitely many directed triangles, then $Z(x, y)$ is infinite. Lemma 2 forces $P(x, y) = \emptyset$. Therefore $J(S)$ is a subgraph of the graph in Lemma 3, proving the degree bound. Lemma 1 gives the final assertion. $\square$

2.4 4. Completing the disproof

Let S be any countably infinite T_* -free tournament.

Case 1: S is nontransitive. By Corollary 4, there is a transitive tournament L on $V(S)$ whose change graph with S is locally finite. Since S is nontransitive, $L \neq S$. Since T_* is nontransitive, L is T_* -free. Take $S' = L$.

Case 2: S is transitive. Reverse any one arc to obtain S' . Every directed triangle in S' must contain that reversed arc.

On the other hand, T_* contains the two edge-disjoint directed triangles

$$c \rightarrow a_0 \rightarrow b_0 \rightarrow c \quad \text{and} \quad c \rightarrow a_1 \rightarrow b_1 \rightarrow c.$$

They share a vertex but no arc. Thus T_* cannot occur in a tournament in which all directed triangles contain one fixed arc. Therefore S' is T_* -free, and its change graph with S consists of one edge.

Both cases produce a distinct T_* -free locally finite perturbation of S . Hence no tournament S_{T_*} as required by the conjecture exists. $\square$

The counterexample addresses the full locally finite perturbation formulation, including its intended nonempty-perturbation convention. There is no remaining gap in this disproof; no assertion is made here about the weaker requirement involving only individual arc reversals.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2506.08810__03
Catalog entry	Conjecture 24
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Infinite induced-saturated graphs
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2506.08810__03/output.md
Independent review	None
arXiv	arXiv:2506.08810
Previous attempt	attacks/2506.08810__03: gpt-5.6-sol verdict partial
Model's caveats	Perturbations are understood to be nonempty; the argument does not settle the weaker single-arc saturation question.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/2506.08810__03/`.